

SHI: Inspecting and Correcting Structured Hypotheses in Interactive Video Retrieval

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Abstract. Large language models can decompose complex video-search narratives into ordered events, but errors in event boundaries or temporal relations may affect downstream retrieval. In multi-event search, such errors are difficult to diagnose when both query interpretation and temporal alignment remain internal to the retrieval pipeline. We present SHI, an interactive video retrieval system with two interaction stages: pre-retrieval and post-retrieval. In the pre-retrieval stage, the Query Hypothesis exposes the LLM-generated decomposition as an editable ordered sequence of events, allowing users to inspect and revise the search structure before retrieval. In the post-retrieval stage, EventTrail exposes the decoded temporal alignment of a candidate video and presents alternative event occurrences as complete ordered paths rather than isolated frames. Users can anchor accepted occurrences or exclude incorrect temporal regions, after which the alignment is re-decoded under the resulting constraints. Because re-decoding operates on retained retrieval evidence, these corrections can be applied without repeating corpus-level multimodal retrieval. Together, the two stages make both semantic decomposition and temporal alignment explicit interaction objects in multi-event video search.

Keywords: Interactive video retrieval · Video Browser Showdown · Temporal alignment · Human-in-the-loop · Multi-event search

1 Introduction

The Video Browser Showdown (VBS) evaluates interactive video retrieval systems through live search tasks over large-scale video collections [17]. Among these tasks, multi-event Known-Item Search (KIS) requires an operator to locate a video described by a sequence of temporally ordered events [17]. Compared with single-event retrieval, this setting introduces two additional sources of uncertainty: the narrative must first be decomposed into suitable retrieval events, and the retrieved evidence must then be aligned to occurrences that satisfy the intended temporal order. Errors in either stage can affect candidate ranking and complicate subsequent inspection. However, these intermediate decisions are often handled internally by the retrieval pipeline, leaving operators to infer how a narrative was interpreted or why particular temporal occurrences were selected from the returned results [1, 8, 15, 3, 6].

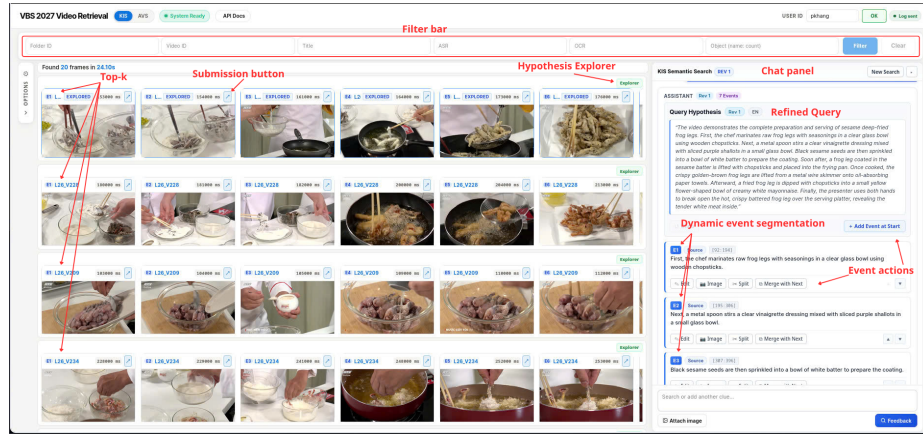


Fig. 1. Overview of the SHI user interface during a multi-event KIS task. The interface integrates candidate video paths with decoded event occurrences and DRES submission (left), the interactive Query Hypothesis editor with dynamic event segmentation and structural actions (right), and the metadata filter bar (top).

LLMs provide a convenient way to convert unstructured narratives into structured event sequences. Prior work on complex task understanding shows that LLMs can identify individual task steps effectively, while still exhibiting difficulty in predicting temporal dependencies between those steps [19]. This distinction is particularly relevant to multi-event video search, where an apparently plausible decomposition may still contain event boundaries or ordering assumptions that do not match the operator’s intended search strategy. If such a decomposition is executed directly, its errors may propagate to multimodal scoring and temporal alignment. We therefore treat the LLM output as a search hypothesis rather than a fixed interpretation and expose it for inspection and editing before retrieval.

Interaction has repeatedly been identified as an important component of VBS search performance. A post-evaluation of VBS 2022 found that strong CLIP-based text retrieval benefits KIS, but needs to be complemented by effective browsing and additional query modalities for consistent performance [14]. More recent interaction logs from VBS 2026 show a strong reliance on iterative, high-frequency text-query reformulation coupled with result-set inspection [4]. Existing interactive video retrieval systems provide rich mechanisms for query formulation, browsing, and relevance feedback [1, 8]; however, for multi-event search, the decoded temporal alignment of a candidate is typically not exposed as an object that the operator can directly inspect and modify. Consequently, when several events are matched correctly but one occurrence is temporally misaligned, the operator may need to reformulate the query or continue browsing despite already having useful retrieval evidence. This motivates interaction not only with the input query, but also with the temporal hypothesis produced for an individual candidate.

To address these two interaction points, we present SHI (Fig. 1), an interactive multi-modal video retrieval system for multi-event KIS with complementary pre-retrieval and

post-retrieval mechanisms. Pre-retrieval Query Hypothesis exposes the LLM-generated event decomposition as an editable structured plan, allowing operators to inspect event boundaries, revise event descriptions, reorder events, and attach visual reference examples before search. Post-retrieval EventTrail exposes the decoded temporal alignment of a candidate video as an ordered event path, allowing operators to inspect alternative occurrences, anchor accepted matches, and exclude incorrect temporal regions. The resulting constraints are applied by re-decoding the alignment from retained retrieval evidence without repeating corpus-level retrieval. Together, these mechanisms make semantic decomposition before retrieval and temporal alignment after retrieval explicit parts of the interactive search process.

2 Related Work

Interactive video retrieval systems developed for the Video Browser Showdown (VBS) combine multimodal retrieval with mechanisms for query formulation, iterative refinement, and result exploration. VISIONE supports textual, visual-example, and sketch-based queries [1], while PraK provides interactive search and filtering over video data services [8]. Exquisitor further combines conversational search with user relevance feedback in a unified retrieval interface [15]. Evaluations of VBS systems have repeatedly highlighted the importance of iterative query reformulation and result inspection in addition to the quality of the initial ranking [7, 14, 4]. These systems provide rich mechanisms for modifying queries and exploring retrieved results, but the decomposition of a narrative request into an ordered sequence of retrieval events is generally not exposed as an editable intermediate representation [1, 8, 15]. SHI addresses this interaction point through the Query Hypothesis, which presents the event decomposition explicitly and allows operators to modify its structure before retrieval.

Temporal structure has also been incorporated into interactive video retrieval in several forms. Vireo-VSS extends its SQL-like interface with templates for specifying temporal query evolution [18], while vitrivr supports multimodal queries with temporal context [3]. More recently, Exquisitor introduced sequence-chain retrieval for temporal queries [6]. Interactive video corpus moment retrieval has also used user feedback to refine the retrieval of temporal intervals [9]. These approaches incorporate temporal structure through query specification, retrieval, or feedback over candidate intervals. SHI focuses on a complementary interaction point: the decoded multi-event alignment of an individual candidate video. Its EventTrail exposes this alignment as an ordered path, allows operators to anchor or reject individual event occurrences, and re-decodes the path under the resulting temporal constraints using retained retrieval evidence.

3 System Overview

SHI separates offline corpus preparation from the online retrieval workflow. During offline preparation, keyframes are enriched with contextual descriptions generated by a vision-language model (Qwen3-VL-8B-Instruct [11]) to form FrameContext metadata. Together with visual embeddings and spoken transcripts, these are compiled into dense FAISS indexes [5] alongside a lexical BM25 index. At runtime, as illustrated in Fig. 2,

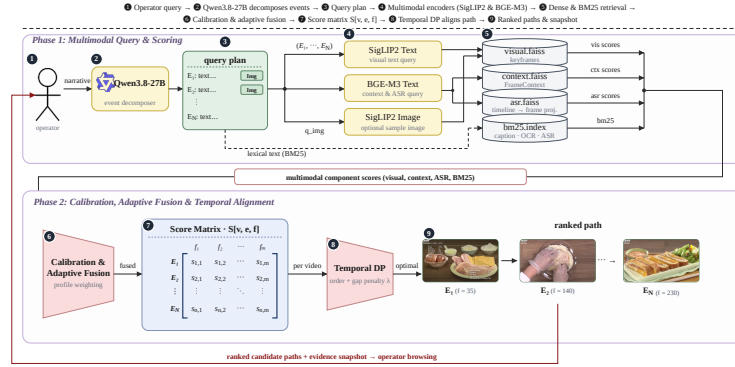


Fig. 2. Multi-event KIS retrieval pipeline in SHI. The Query Hypothesis provides an ordered sequence of events. Text events query visual, FrameContext, ASR, and lexical indexes, while optional reference images contribute visual similarity scores. Component scores are calibrated and fused into an event–frame score matrix $S[v, e, f]$ for each candidate video v . A dynamic-programming decoder then extracts chronologically ordered event paths, which are used to rank candidate videos and construct the evidence snapshot.

these artifacts are loaded as read-only retrieval structures and queried for each event in the active Query Hypothesis. Corpus-side feature extraction and index construction therefore remain outside the interactive request path, while online processing is limited to query encoding, retrieval, score fusion, and temporal alignment.

3.1 Multi-event KIS Retrieval Pipeline

Multi-event KIS is formulated as multimodal event scoring followed by joint temporal alignment. Given an ordered sequence of events from the Query Hypothesis, each textual event is evaluated against several retrieval sources: SigLIP2 [16] provides visual text–frame similarity, BGE-M3 [2] provides dense FrameContext and ASR evidence, and BM25 [12] provides lexical matching over frame-associated text. Optional reference images contribute additional SigLIP2 image–frame similarity scores. The resulting components are calibrated and combined by event-adaptive weighted fusion into an event–frame score matrix $S[v, e, f]$ for each candidate video v . A dynamic-programming decoder then selects one canonical frame occurrence for each event subject to strict temporal ordering. Its objective combines the fused event–frame scores with a temporal gap penalty that discourages large separations between consecutive event occurrences. Candidate videos are ranked by the scores of their decoded paths, so the ranking accounts jointly for event-level evidence and temporal ordering rather than aggregating independently selected frame matches.

3.2 Evidence Snapshots and Competition Integration

To support post-retrieval interaction without repeating corpus-level search, SHI captures an in-memory Evidence Snapshot after the initial KIS execution. For the returned

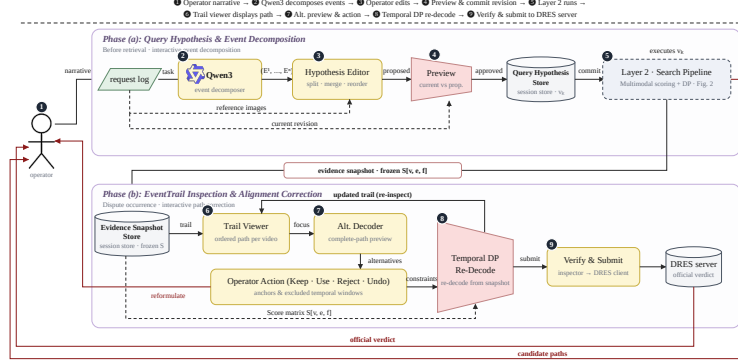


Fig. 3. The two hypothesis-level interaction mechanisms in SHI. (a) Query Hypothesis exposes the initial LLM-derived event decomposition before retrieval and supports structural editing, reference-image binding, and non-destructive preview before a revision is committed. (b) Event-Trail exposes the decoded temporal path of a candidate video and presents temporally distinct alternatives as complete ordered paths. Keep, Use, and Reject actions introduce anchor or exclusion constraints, after which the path is re-decoded from the retained evidence snapshot.

candidates, the snapshot retains immutable event-frame evidence together with the event identifiers, initial decoded paths, and the decoder configuration used for that search. When an operator applies an EventTrail action, such as anchoring an accepted occurrence or rejecting a temporal region, the corresponding constraints are maintained separately from the frozen evidence and supplied to the temporal decoder. The decoder then re-computes the feasible path for the selected video using the retained evidence, without repeating query embedding or corpus index search. Once the operator selects a frame for submission, SHI’s integrated competition client converts the selection to the required submission format and sends it to the Distributed Retrieval Evaluation Server (DRES) [13], reducing the need for manual transfer of frame coordinates during live search.

4 Interactive Hypothesis Inspection and Path Alignment

SHI exposes two intermediate representations for direct interaction during multi-event search. In the pre-retrieval stage, the Query Hypothesis represents the current interpretation of a narrative as an ordered sequence of events that can be inspected and edited before search. In the post-retrieval stage, EventTrail represents the temporal alignment of an individual candidate video as an ordered path over event occurrences. As illustrated in Fig. 3, edits to the Query Hypothesis determine the event specifications used by retrieval, whereas EventTrail actions are translated into temporal constraints and re-decoded over retained retrieval evidence. The two stages therefore expose semantic decomposition and temporal alignment as separate but connected interaction points in the retrieval workflow.

4.1 Pre-retrieval Query Hypothesis

The Query Hypothesis represents the current semantic interpretation of a multi-event request before retrieval is executed. Given a narrative query, a large language model (Qwen3 [10]) proposes an ordered sequence of events $(E_1, \dots, E_N)$. For the initial decomposition, each proposed event must correspond to a verbatim fragment of the original query; the server verifies these fragments against the source text and falls back to a single-event representation if the decomposition cannot be grounded. The resulting event sequence is shown explicitly to the operator together with its temporal order and any event-specific reference images. Rather than treating the decomposition as a fixed parser output, SHI keeps it as the active search hypothesis from which the downstream retrieval plan is constructed.

The editor supports event-level structural operations including splitting and merging events, changing their order, editing or adding event descriptions, and attaching or removing reference images. Operators can also provide conversational feedback to prompt Qwen3 for targeted hypothesis revisions. Each proposed operation can first be previewed without modifying the active hypothesis, allowing the operator to compare the current and proposed event sequences before committing the change. A committed operation creates a new monotonically revisioned hypothesis and stores the previous state in a bounded session history, from which the latest change can be undone. Subsequent retrieval uses the committed revision, so structural edits directly change the event specifications supplied to the multimodal scoring and temporal-alignment pipeline.

4.2 Post-retrieval EventTrail

EventTrail operates on the decoded temporal path of an individual candidate video after the initial retrieval. For a query with N events, the active trail contains one selected frame occurrence for each event in temporal order. When an operator focuses on event E_e , SHI searches for alternative occurrences by evaluating candidate frame positions for that event in the context of the complete event sequence, rather than ranking frames only by their local event score $S[v, e, f]$.

Let $P_e(f)$ denote the score of the highest-scoring valid prefix path ending with event E_e at frame f , and let $Q_e(f)$ denote the score of the highest-scoring valid suffix path starting from the same assignment. The score of the best complete path constrained to pass through frame f at event E_e is

$$C_e(f) = P_e(f) + Q_e(f) - S[v, e, f]. \quad (1)$$

The subtraction removes the event-frame score $S[v, e, f]$, which is included in both the forward and backward terms. Thus, $C_e(f)$ evaluates an occurrence according to the best complete chronological path that contains it, rather than according to the frame score in isolation. Candidate frames are considered in descending order of $C_e(f)$. To avoid presenting several alternatives from the same temporal neighborhood, SHI retains a candidate only if its timestamp is at least Δ_{alt} away from every previously retained representative. The process stops after K_{alt} alternatives have been selected. In the current implementation, $K_{\text{alt}} = 4$ and $\Delta_{\text{alt}} = 5$ seconds. Each retained representative is presented

together with its complete decoded path, allowing the operator to inspect how choosing that occurrence affects the remaining events before committing a change.

For each retained representative at timestamp t_j , EventTrail constructs a bounded occurrence interval I_j around that representative. Ignoring clipping at the active video boundaries, its radius is

$$r_j = \min \left(r_{\max}, \frac{1}{2} \min_{k \neq j} |t_j - t_k| \right), \quad (2)$$

where $r_{\max} = 8$ seconds in the current implementation. If no competing representative exists, the implementation defines the empty minimum as $+\infty$, so that $r_j = r_{\max}$. Representatives are processed in chronological order (with their original order used as a stable tie-breaker); neighboring intervals are then split at their temporal midpoint so that they do not overlap, and the resulting interval is clipped to the active video boundaries. Alternative representatives are separated by $\Delta_{\text{alt}} = 5$ seconds during candidate generation, so equal timestamps are not normally returned as distinct alternatives. The interval therefore represents the temporal neighborhood associated with an alternative occurrence and defines the region affected by rejection.

The operator can apply three occurrence-level actions. Keep anchors the frame currently assigned to the focused event. Use anchors the representative frame of a selected alternative and re-decodes the complete path under that anchor. Reject adds the corresponding interval I_j to the exclusion set of the focused event only. Occurrences of the remaining events are not directly excluded, although their selected frames may change when the complete path is re-decoded under the updated temporal constraints. All actions leave the underlying Evidence Snapshot unchanged, so re-decoding reuses the retained event-frame evidence without repeating query embedding or corpus-level retrieval. EventTrail also maintains a bounded history of constraint states, allowing the most recent committed action to be undone before further inspection or submission.

5 Conclusion

We presented SHI, an interactive multimodal video retrieval system for multi-event KIS at VBS 2027. SHI exposes two intermediate retrieval hypotheses for direct user interaction. The pre-retrieval Query Hypothesis represents the LLM-generated event decomposition as an editable ordered sequence, while the post-retrieval EventTrail exposes the decoded temporal alignment of an individual candidate video and supports occurrence-level correction through anchor and exclusion constraints. By retaining event-frame evidence in session-scoped Evidence Snapshots, constrained paths can be re-decoded without repeating corpus-level retrieval. Together, these mechanisms make semantic decomposition before retrieval and temporal alignment after retrieval explicit interaction points in multi-event search.

References

1. Amato, G., Bolettieri, P., Carrara, F., Falchi, F., Gennaro, C., Messina, N., Vadicamo, L., Vairo, C.: VISIONE 5.0: Enhanced user interface and AI models for VBS2024. In: MultiMe-

- dia Modeling. Lecture Notes in Computer Science, vol. 14557, pp. 332–339. Springer (2024). https://doi.org/10.1007/978-3-031-53302-0_29
2. Chen, J., Xiao, S., Zhang, P., Luo, K., Lian, D., Liu, Z.: M3-embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. arXiv preprint arXiv:2402.03216 (2024)
3. Heller, S., Arnold, R., Gasser, R., Gsteiger, V., Parian-Scherb, M., Rossetto, L., Sauter, L., Spiess, F., Schuldt, H.: Multi-modal interactive video retrieval with temporal queries. In: MultiMedia Modeling. Lecture Notes in Computer Science, vol. 13142, pp. 493–498. Springer (2022). https://doi.org/10.1007/978-3-030-98355-0_44
4. Jäckl, B., Khan, O.S., Verner, B.V., Vopalkova, Z., Schlegel, U., Keim, D.A., Lokoč, J.: What drove success at the 15th video browser showdown? a comprehensive interaction-logging analysis. In: Proceedings of the 2026 International Conference on Multimedia Retrieval. pp. 1730–1739. ACM (2026). <https://doi.org/10.1145/3805622.3810635>
5. Johnson, J., Douze, M., Jégou, H.: Billion-scale similarity search with GPUs. IEEE Transactions on Big Data 7(3), 535–547 (2019). <https://doi.org/10.1109/TBDATA.2019.2921572>
6. Khan, O.S., Sharma, U., Marcelino, G., Rudinac, S., Jónsson, B.P.: Exquisitor at the video browser showdown 2026: Temporal queries revisited. In: MultiMedia Modeling. Lecture Notes in Computer Science, vol. 16415, pp. 245–251. Springer (2026). https://doi.org/10.1007/978-981-95-6963-2_27
7. Lokoč, J., Vopálková, Z., Dokoupil, P., Peška, L.: Video search with CLIP and interactive text query reformulation. In: MultiMedia Modeling. Lecture Notes in Computer Science, vol. 13833, pp. 628–633. Springer (2023). https://doi.org/10.1007/978-3-031-27077-2_50
8. Lokoč, J., Vopálková, Z., Stroh, M., Buchmüller, R., Schlegel, U.: PraK tool: An interactive search tool based on video data services. In: MultiMedia Modeling. Lecture Notes in Computer Science, vol. 14557, pp. 340–346. Springer (2024). https://doi.org/10.1007/978-3-031-53302-0_30
9. Ma, Z., Ngo, C.W.: Interactive video corpus moment retrieval using reinforcement learning. In: Proceedings of the 30th ACM International Conference on Multimedia. pp. 296–306. ACM (2022). <https://doi.org/10.1145/3503161.3548277>
10. Qwen Team: Qwen3 technical report (2025), <https://arxiv.org/abs/2505.09388>
11. Qwen Team: Qwen3-VL technical report. arXiv preprint arXiv:2511.21631 (2025)
12. Robertson, S., Zaragoza, H.: The probabilistic relevance framework: BM25 and beyond. Foundations and Trends in Information Retrieval 3(4), 333–389 (2009). <https://doi.org/10.1561/15000000019>
13. Rossetto, L., Gasser, R., Sauter, L., Bernstein, A., Schuldt, H.: A system for interactive multimedia retrieval evaluations. In: MultiMedia Modeling. Lecture Notes in Computer Science, vol. 12573, pp. 385–390. Springer (2021). https://doi.org/10.1007/978-3-030-67835-7_33
14. Schall, K., Bailer, W., Barthel, K.U., Carrara, F., Lokoč, J., Peška, L., Schoeffmann, K., Vadicamo, L., Vairo, C.: Interactive multimodal video search: An extended post-evaluation for the VBS 2022 competition. International Journal of Multimedia Information Retrieval 13(2), 15 (2024). <https://doi.org/10.1007/s13735-024-00325-9>
15. Sharma, U., Khan, O.S., Rudinac, S., Jónsson, B.P.: Exquisitor at the video browser showdown 2025: Unifying conversational search and user relevance feedback. In: MultiMedia Modeling. Lecture Notes in Computer Science, vol. 15524, pp. 264–271. Springer (2025). https://doi.org/10.1007/978-981-96-2074-6_31
16. Tschannen, M., Gritsenko, A., Wang, X., Naeem, M.F., Alabdulmohsin, I., Parthasarathy, N., Evans, T., Beyer, L., Xia, Y., Mustafa, B., Hénaff, O., Harmsen, J., Steiner, A., Zhai, X.: SigLIP 2: Multilingual vision-language encoders with improved semantic understanding, localization, and dense features. arXiv preprint arXiv:2502.14786 (2025)
17. Video Browser Showdown Organizers: Call for papers (VBS 2027). <https://videobrowsershowdown.org/call-for-papers/>, accessed 21 September 2026
18. Wu, J., Nguyen, P.A., Ma, Z., Ngo, C.W.: SQL-like interpretable interactive video search. In: MultiMedia Modeling. Lecture Notes in Computer Science, vol. 12573, pp. 391–397. Springer (2021). https://doi.org/10.1007/978-3-030-67835-7_34
19. Yuan, Q., Kazemi, M., Xu, X., Noble, I., Imbrasaitė, V., Ramachandran, D.: Tasklama: Probing the complex task understanding of language models. In: Proceedings of the AAAI Conference on Artificial Intelligence (2024). <https://doi.org/10.1609/aaai.v38i17.29918>